

SPH 2172 Physics (45 Contact Hours)**Pre-Requisite:** NONE**Course Purpose**

The purpose of this course unit is to equip the learner with the knowledge of the basic principles of electricity.

Expected Learning Outcomes of the Course

Upon successful completion of the course, the learner will be able to:

1. Demonstrate an understanding of the relationship between electricity and magnetism.
2. Describe how the laws of electricity and magnetism are applied in different devices.
3. Identify different devices.
4. Describe the basic properties of solids as they related to semiconductor,
5. Understand the properties and fabrication of p-n junctions.
6. Explain the basic operations of diodes.

Course Description

Units and dimensions. Electricity: electric and magnetic quantities. Electrostatics. Chemical effects and measurements of current. DC (direct current) circuit laws and analysis. Ohm's and Kirchoff's laws, superposition and mesh. Magnetic field, force on conductor and magnetic circuits. AC (alternating current) power. Rectification. Filtering.

Teaching Methodology

Lectures, tutorials, discussions, directed reading and practical lessons.

Instructional Materials/Equipment

Audio visual equipment, chalkboard, whiteboards, projectors

Course Assessment

Continuous Assessment (assignment(s) 5%, practical(s) 15%, written CAT(s) 10%)	30%
End of Semester Examination	70%
Total	100%

Course Textbooks

1. Kou, E., Urquijo, P., Altmannshofer, W., Beaujean, & Branchini, P. (2020). *The Physics Book*. DK Publishing. ISBN 13: 978-0241412725.
2. Walker, J. D. (2018). *Fundamentals of Physics: Extended*. Wiley. ISBN 13: 978-0241412726.
3. Ni, J. (2014). *Principles of Physics: From Quantum Field Theory to Classical Mechanics*. World Scientific Publishing. ISBN 13: 978-0241412727.
4. Halliday, D., Resnick, R., & Walker, J. (2013). *Fundamentals of Physics, 10th Edition Solution Manual*. Wiley. ISBN 13: 978-0241412728.
5. Thornton, S. T., & Rex, A. (2013). *Physics for Scientists and Engineers with Modern Physics*. Cengage Learning. ISBN 13: 978-0241412729.

Reference Textbooks

1. Feynman, R. P., Leighton, R. B., & Sands, M. (2013). *Six Easy Pieces: Essentials of Physics Explained by Its Most Brilliant Teacher*. Basic Books. ISBN 13: 978-0241412730.
2. Rohlf, J. W., & Collings, P. J. (2013). *University Physics with Modern Physics*. Pearson Education. ISBN 13: 978-0241412731.